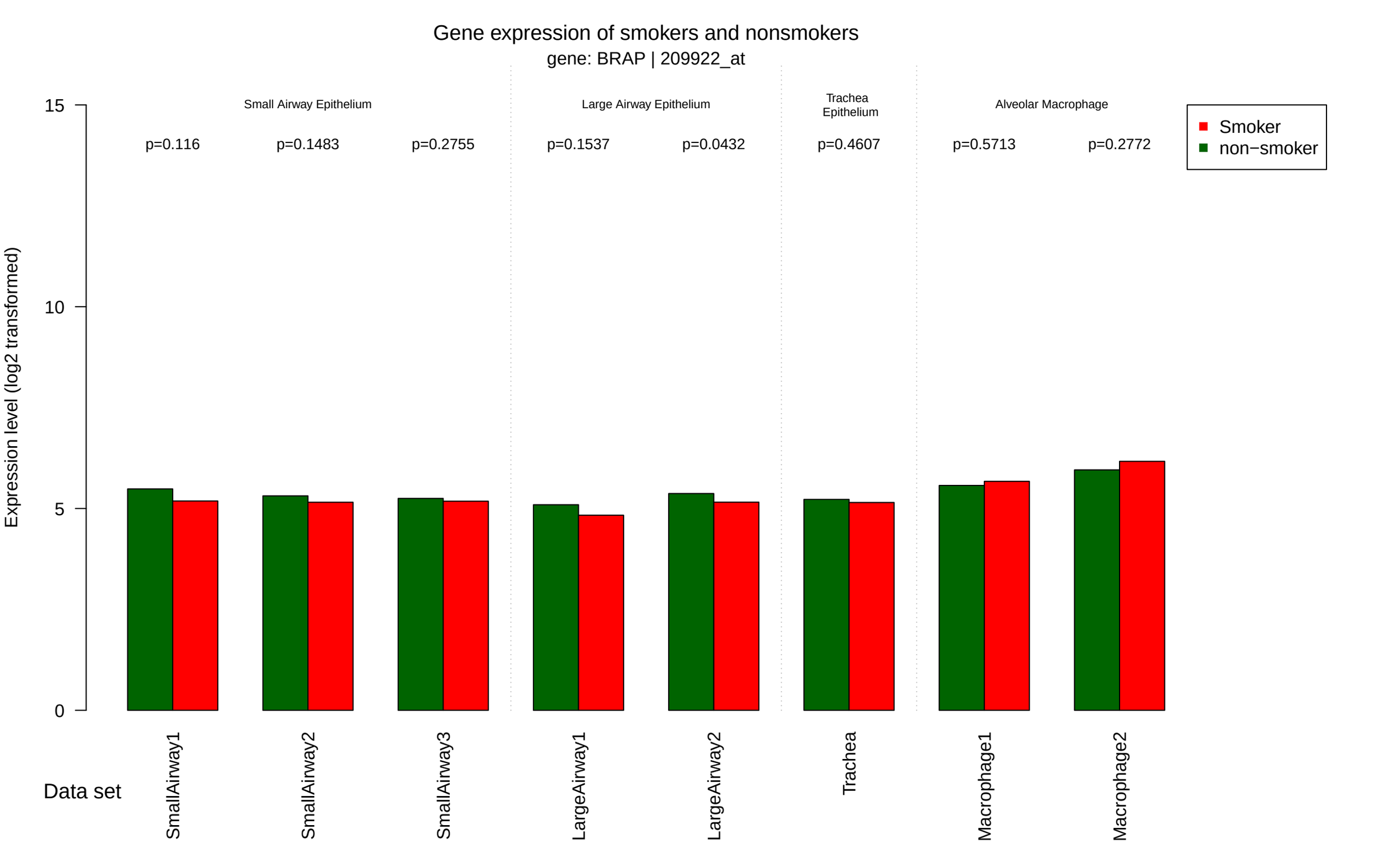
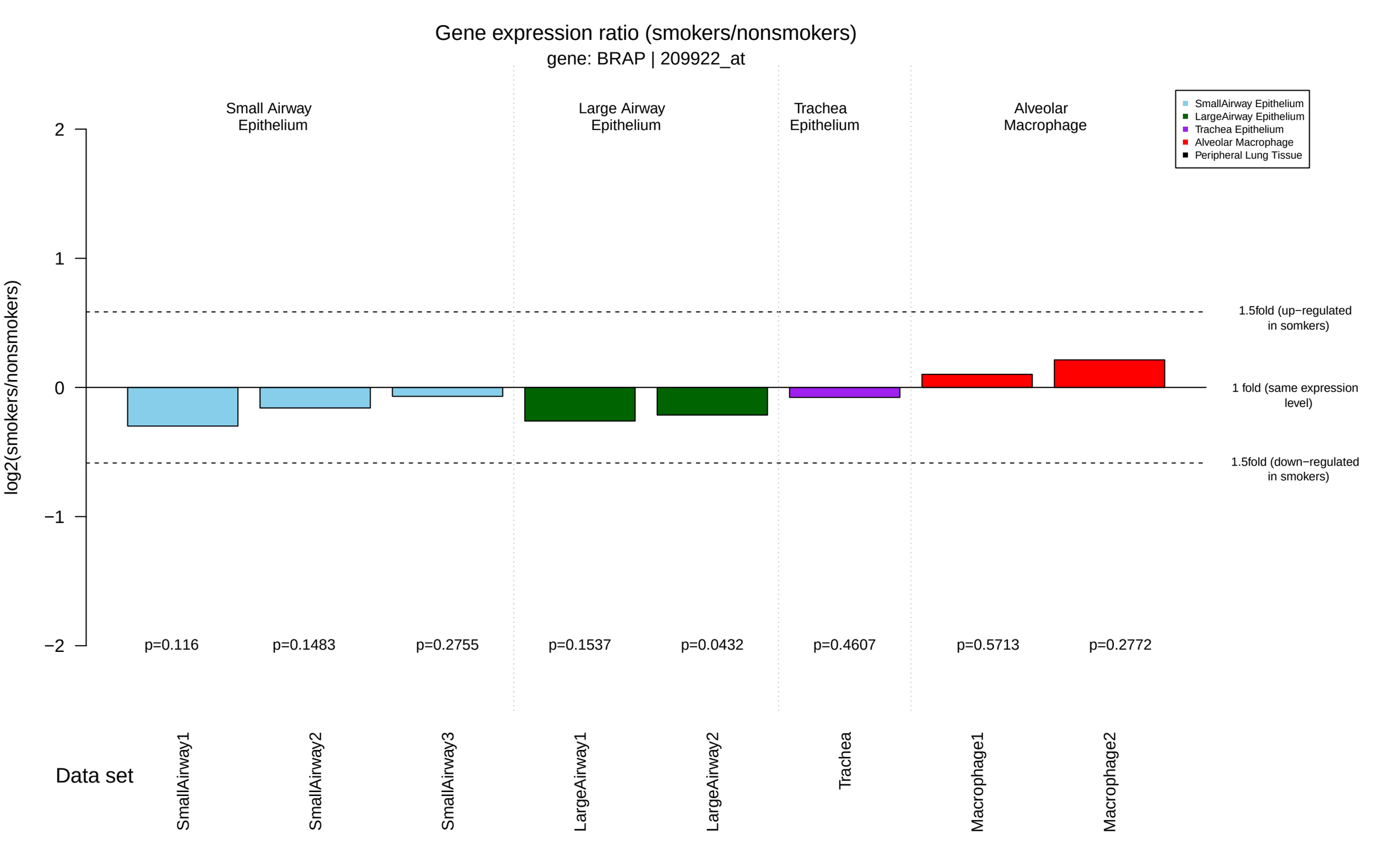
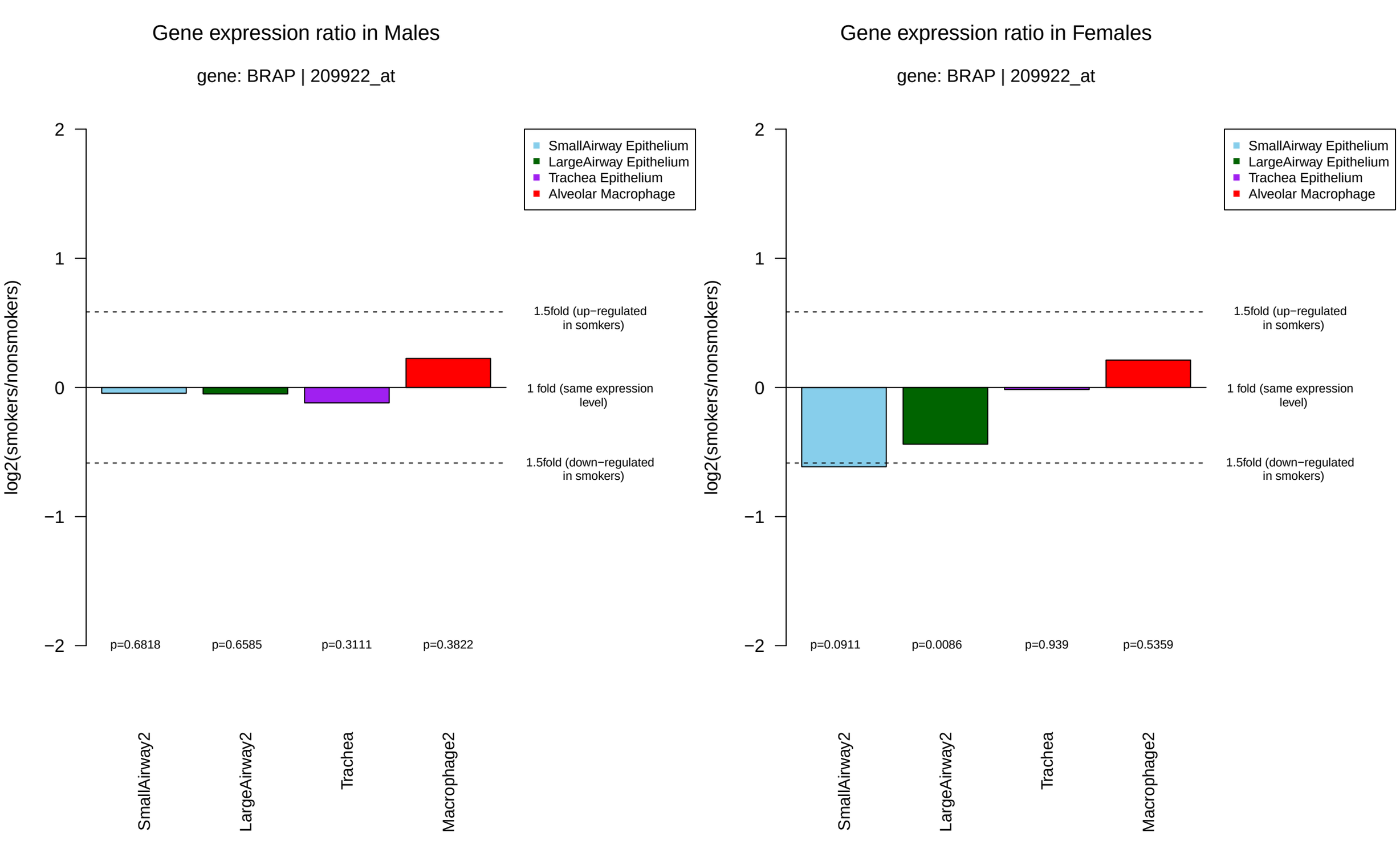
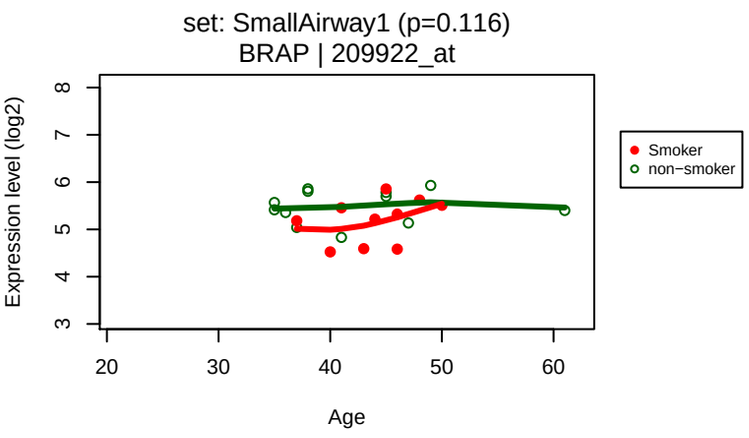
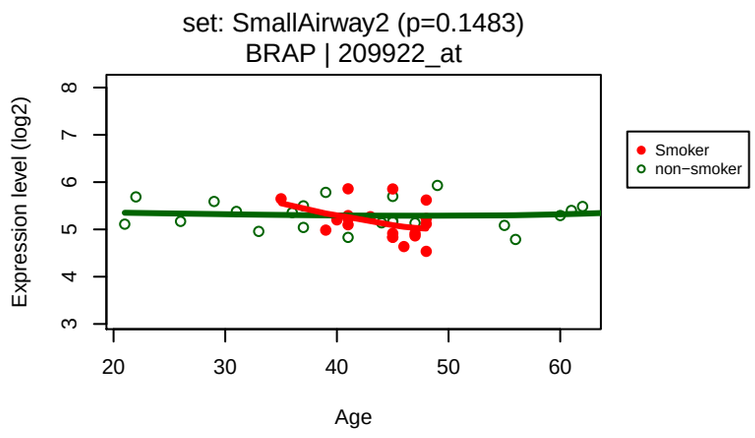
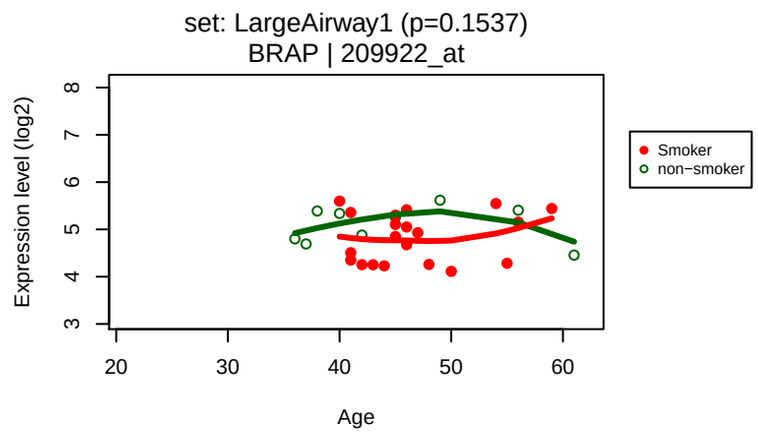

Gene expression of smokers and nonsmokers
gene: BRAP | 209922_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.116
p=0.1483
p=0.2755
p=0.1537
p=0.0432
p=0.4607
p=0.5713
p=0.2772
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

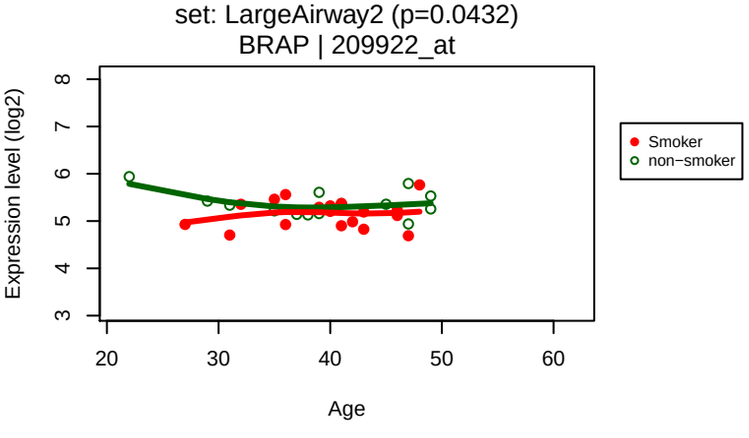

Gene expression ratio (smokers/nonsmokers)
gene: BRAP | 209922_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.116
p=0.1483
p=0.2755
p=0.1537
p=0.0432
p=0.4607
p=0.5713
p=0.2772
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

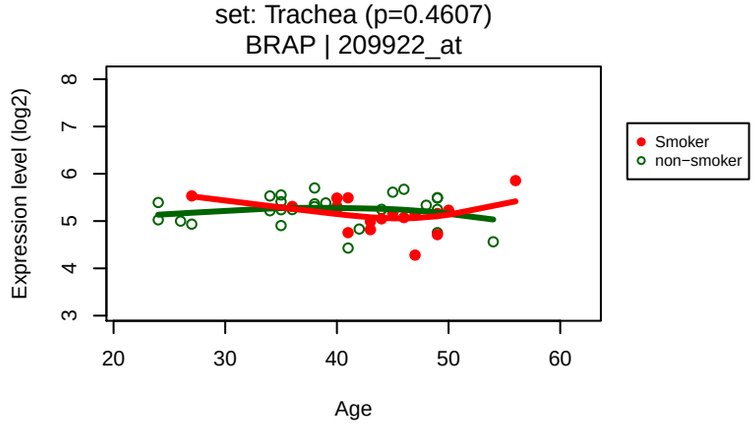

Gene expression ratio in Males
gene: BRAP | 209922_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6818
p=0.6585
p=0.3111
p=0.3822
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRAP | 209922_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0911
p=0.0086
p=0.939
p=0.5359
SmallAirway2
LargeAirway2
Trachea
Macrophage2

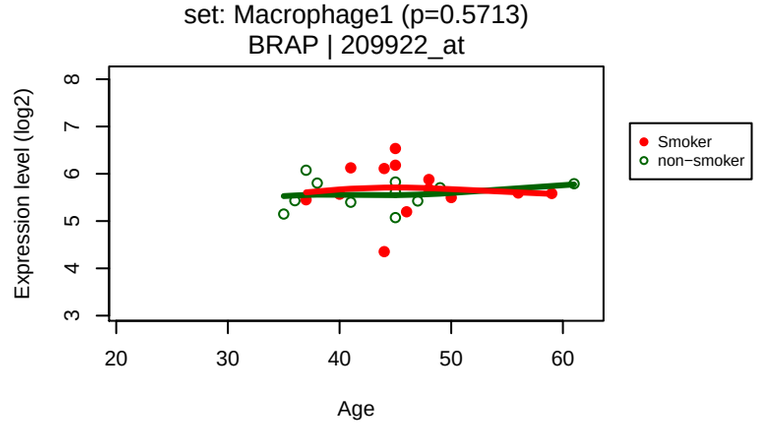

set: SmallAirway1 (p=0.116)
BRAP | 209922_at
Expression level (log2)
Age
Smoker
non−smoker

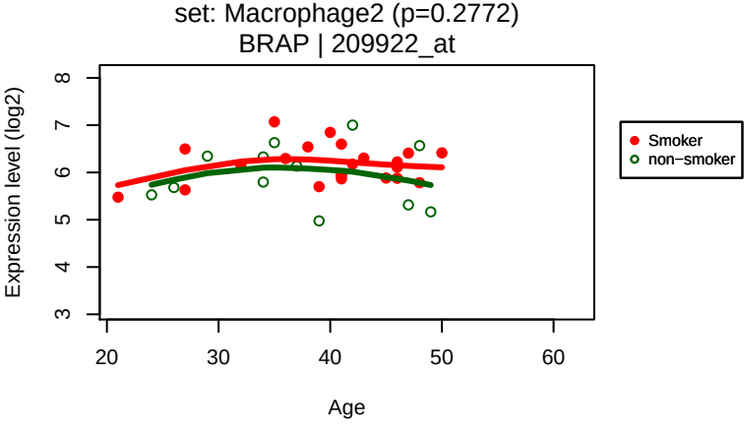

set: SmallAirway2 (p=0.1483)
BRAP | 209922_at
Expression level (log2)
Age
Smoker
non−smoker

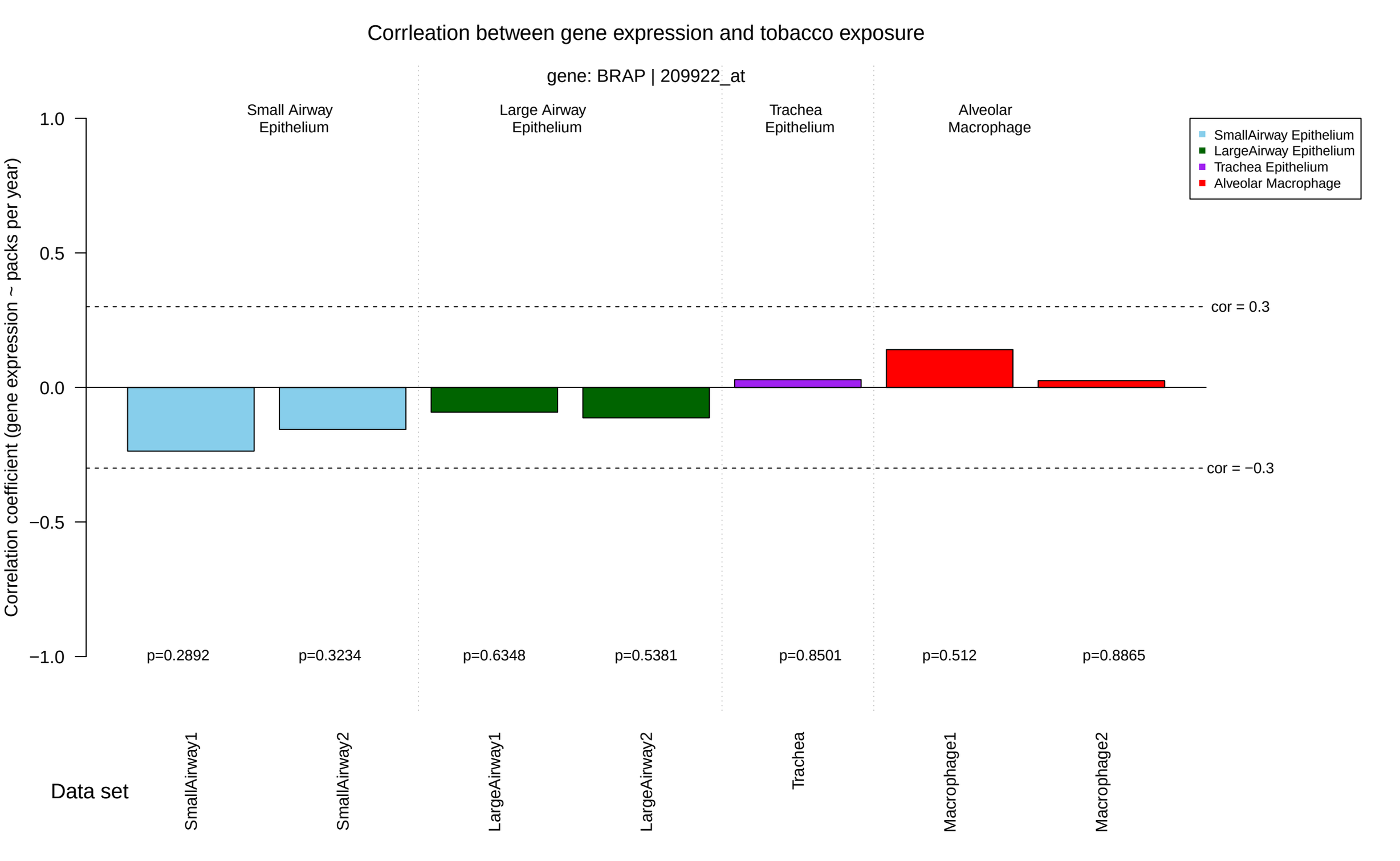

set: LargeAirway1 (p=0.1537)
BRAP | 209922_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0432)
BRAP | 209922_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4607)
BRAP | 209922_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5713)
BRAP | 209922_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2772)
BRAP | 209922_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRAP | 209922_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2892
p=0.3234
p=0.6348
p=0.5381
p=0.8501
p=0.512
p=0.8865
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2